INDUSTRIAL HYGIENE & CHEMICAL SAFETY

1

Session Objectives

At the end of the session, participants will be able to:

- Explain the **principles** of Industrial Hygiene
- **Identify health hazards** in the workplace
- Discuss the concepts of a chemical safety program focusing on **proper chemical labelling** and **storage**.

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What is Industrial Hygiene ?

The science and art devoted to *identification, evaluation* and *control* of environmental factors and stresses arising in or from the workplace, which may cause sickness, impaired health and well-being, or significant discomfort among workers or among citizens of the community.

3

Health Hazards Classification

Chemical

Physical

Biological

Ergonomic Hazards

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Chemical Hazards

These arise from excessive airborne concentration of …

- Vapors
- Mists
- Fumes
- Gases
- Dusts

5

Chemical Hazards

Q: How do chemicals become a hazard?

A: When they become airborne and the concentration is excessive…

Chemical Hazards	Nature / Form
Organic Solvents	Vapor
Acids / Bases	Mists
Dust / Particulates	Powder / Fibers
Heavy Metal	Fumes
Gases	Gas

6

Physical Hazards

- Noise
- Vibration
- Illumination
- Extreme temperature
- Extreme pressure
- Radiation

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Physical Hazards…

Noise

- **Unwanted and excessive sound**

  - A form of energy caused by the vibration of air

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Arm's Length Rule

"If two (2) people with no hearing impairment have to raise their voices or shout to be heard in a distance of less than arms length from each other, the sound level is potentially hazardous."

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Physical Hazards…

Vibration

- **is a physical factor that acts on man by transmission of mechanical energy from sources of oscillation.**
  - ***Low frequency or whole body***
  - ***High frequency or segmental***
    - **– hand driven power tools such as portable grinder and polishers**

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Physical Hazards…

Illumination

➢ is the measure of stream of light falling on a surface

Natural

Artificial

❖ Types of Workplace Lighting

1. General lighting
2. Local lighting

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Physical Hazards…

Extreme Temperature

Factors affecting Heat Exposure

Thermal factors

➢ temperature & humidity

Physical Workload

➢ light, moderate, heavy & very heavy

Work-Rest Regimen

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Biological Hazards

1. Microbiological

- ☑ Bacteria, viruses, molds, fungi and protozoa

2. Macrobiological

- ☑ Insects, parasites, plants and animals

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Ergonomic Hazards

Ergonomics

- Greek derivation: Ergo (work)
  Nomos (law)

- "Fitting the TASK to the HUMAN"

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Ergonomic Hazard Examples

- Improperly designed tools or work areas
- Improper lifting or reaching
- Poor visual conditions
- Repeated motion in awkward position

15

Summary of Health Hazards

Physical
noise, vibration, radiation, defective illumination, temperature extremes

Chemical
dusts, gases, vapors, fumes, mists, etc.

Biological
viruses, bacteria, fungi, parasites, insects, etc.

Ergonomic
exhaustive physical exertions, excessive standing, improper motions, lifting heavy load, job monotony, workplace stress, etc.

16

How HAZARDS are IDENTIFIED?

- Walk through survey / ocular inspection
- Review of processes involved
- Gathering workers' complaints
- Knowing the raw materials used, products and by-products
- GHS Labels and Safety Data Sheet

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What is a Label?

An appropriate group of **written**, **printed** or **graphic information** elements that are affixed to, printed on, or **attached** to the **immediate container** of a **hazardous product**, or to the outside packaging of a hazardous product.

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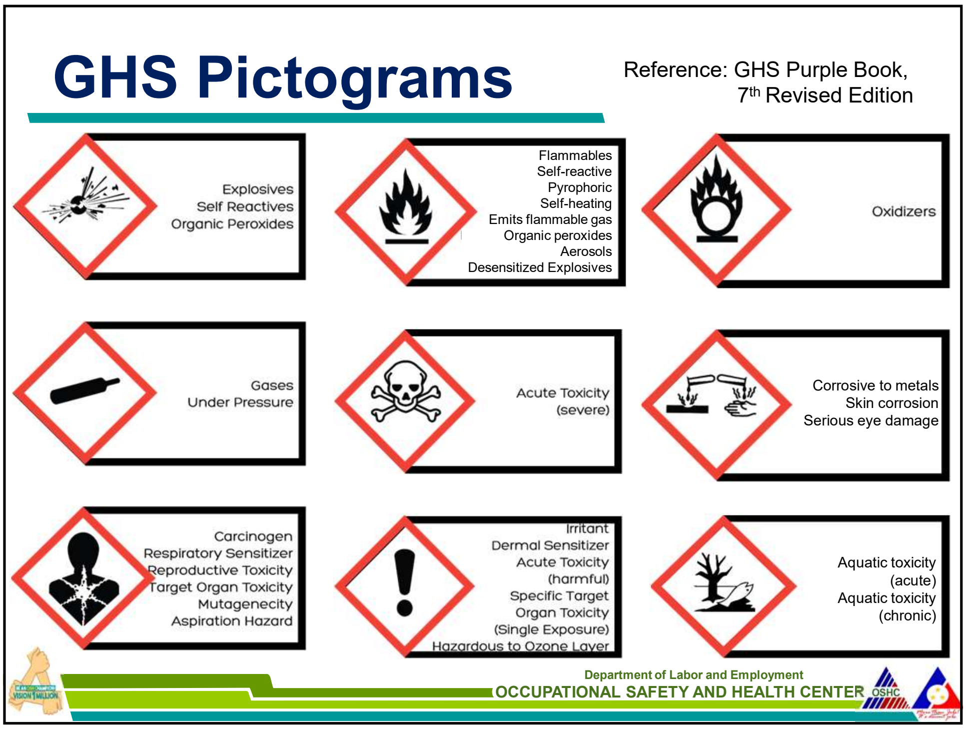
Components Of A GHS-Compliant Label
product identifier
AMMONIA
signal word
DANGER
hazard statement
TOXIC IF INGESTED
pictograms
Wash hands thoroughly after handling. Keep container tightly closed when not in use Keep away from heat, sparks and open flames - may explode when exposed to high heat. Use in an open area that is well-ventilated. Breathing in ammonia is irritating and corrosive. Wear protective gloves and safety goggles to prevent burns and irritation.
precautionary statements
If swallowed: Immediately call Poison Control or doctor/physician. Drink water or milk to dilute ammonia.
See Safety Data Sheet (SDS) for further details regarding safe use of this product.
supplier information
ABC Chemicals - 123 Main Street - Cincinnati, OH - www.abcchem.com - 800-733-5252

19


GHS Pictograms
Reference: GHS Purple Book, 7th Revised Edition
Explosives
Self Reactives
Organic Peroxides
Flammables
Self-reactive
Pyrophoric
Self-heating
Emits flammable gas
Organic peroxides
Aerosols
Desensitized Explosives
Oxidizers
Gases
Under Pressure
Acute Toxicity
(severe)
Corrosive to metals
Skin corrosion
Serious eye damage
Carcinogen
Respiratory Sensitizer
Reproductive Toxicity
Target Organ Toxicity
Mutagenecity
Aspiration Hazard
Irritant
Dermal Sensitizer
Acute Toxicity
(harmful)
Specific Target
Organ Toxicity
(Single Exposure)
Hazardous to Ozone Layer
Aquatic toxicity
(acute)
Aquatic toxicity
(chronic)
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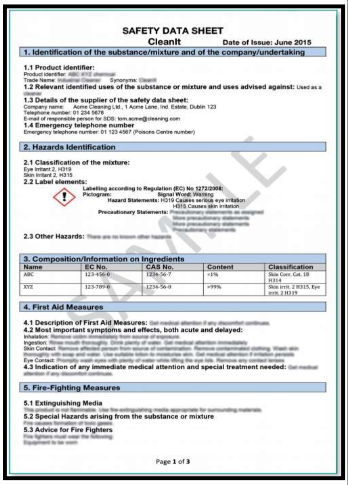
20

What is an SDS?

Safety Data Sheet

A summary of the important **health**, **safety** and **toxicological information** on the **chemical** or the mixture ingredients.



SAFETY DATA SHEET

CleanIt Date of Issue: June 2015

1. Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier:
Product identifier:
Trade Name: Synonyms:
1.2 Relevant identified uses of the substance or mixture and uses advised against: Used as a
1.3 Details of the supplier of the safety data sheet:
Company name: Acme Cleaning Ltd., 1 Acme Lane, Ind. Estate, Dublin 123
Telephone number: 01 234 5678
E-mail of responsible person for SDS: tom.acme@cleaning.com
1.4 Emergency telephone number
Emergency telephone number: 01 123 4567 (Poisons Centre number)

2. Hazards Identification

2.1 Classification of the mixture:
Eye Irritant 2, H319
Skin Irritant 2, H315
2.2 Label elements:
Labelling according to Regulation (EC) No 1272/2008:
Pictogram: Signal Word: Warning
Hazard Statements: H319 Causes serious eye irritation
H315 Causes skin irritation
Precautionary Statements:
2.3 Other Hazards:

3. Composition/Information on Ingredients

Name	EC No.	CAS No.	Content	Classification
ABC	123-456-0	1234-56-7	<1%	Skin Corr. Cat. 1B H314
XYZ	123-789-0	1234-56-0	>99%	Skin irrit. 2 H315, Eye irrit. 2 H319

4. First Aid Measures

4.1 Description of First Aid Measures:
4.2 Most important symptoms and effects, both acute and delayed:
Inhalation:
Ingestion:
Skin Contact:
Eye Contact:
4.3 Indication of any immediate medical attention and special treatment needed:

5. Fire-Fighting Measures

5.1 Extinguishing Media
5.2 Special Hazards arising from the substance or mixture
5.3 Advice for Fire Fighters

Page 1 of 3

Importance of SDS

- Primarily for workplace use
- Employees & workers use these as a source of **information** and **safety precautions** about the **chemicals** and their **hazards**

Contents of an SDS

1. Identification of the Chemicals
2. Hazards identification
3. Composition / Information on ingredients
4. First-aid measures
5. Fire fighting measures
6. Accidental release measures
7. Handling and storage
8. Exposure control and personal protection
9. Physical and chemical properties
10. Stability and reactivity
11. Toxicological information
12. Ecological information
13. Disposal considerations
14. Transport information
15. Regulatory information
16. Other information

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DOLE Department Order No. 136-14

"Guidelines for the Implementation of the Globally Harmonized System in Chemical Safety Program in the Workplace"

Section 1

- This guideline shall **apply** to all **workplaces** engaged in the **manufacture**, **use**, **storage** of **industrial chemicals**, in the private sector, including their supply chain.

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DOLE Department Order No. 136-14

Section 2

This guideline aims to **protect workers** and **properties** from the **hazard** of **chemicals** and to **prevent** or **reduce** the incidence of **chemically-induced accidents**, **illnesses** and **injuries** and **death** resulting in the use of chemicals at work.

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General Considerations for Chemical Storage

1. Chemical Identification
2. Separation and Isolation
3. Storage Containers
4. Ventilation
5. Access and Egress

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What do we need to know on safe handling of chemicals?

- **Safety** user is potentially exposed to the chemical.
- **Risk / Consequences** due to lack of safety considerations in the use of chemicals.
- **Where** are the chemicals used and handled?
- **Frequency / Quantity** of chemical used?
- **How** was the chemical used?

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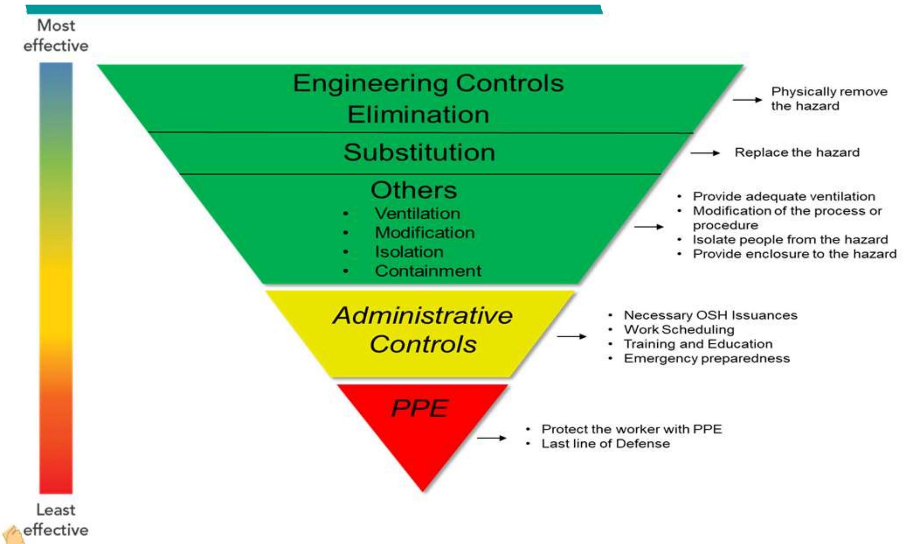
Hierarchy of Controls



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Common Code of Practice at the Workplace

- Make an **inventory list** of the chemicals used.
- Make a **summary** of the health effects of those chemicals.
- Specify the **route of exposure** of those chemicals.
- Check and use **necessary tools or equipment** for safe work practices.

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Common Code of Practice at the Workplace

- Make a breakdown of **operating procedures** and **control measures** in cases of emergency situations (chemical spills)
- **Train workers** with the proper use and safe handling of chemicals
- Check for the availability of **PPE**

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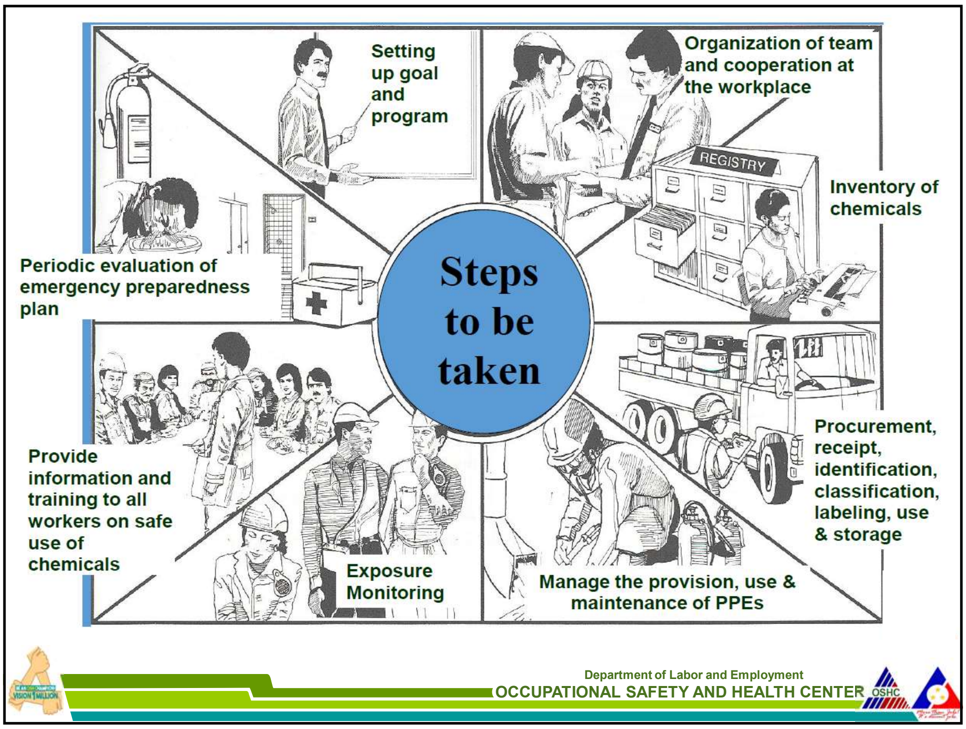
30



31

Review

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Review Questions

- What do we call the document that contains information on the chemicals that we use?
- Where should it be located?
- How does a GHS Pictogram look like?

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Clear Points

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- Chemicals should be stored and handled properly to prevent the damage to property, environment and to human lives.
- Every container must have appropriate labels and pictograms for proper identification and use.
- We can refer to Safety Data Sheet on the health effects of chemicals used in my workplace.

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35

Thank you for listening

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36